

ECOM EXPERTS

Ecom revenue engine

Manual de usuario, técnico y de desarrollo

ecom_revenue_engine · 08/09/2026

Contenido

1. Introduction

1.1 Key Features

2. System Requirements

2.1 Compatibility

2.2 No External Dependencies

2.3 AI Provider Requirements

3. Installation

3.1 Standard Installation

3.2 Post-Installation Steps

3.3 Upgrading

4. Administration Panel

4.1 Interfaz Administrativa Profesional (v1.1.0)

4.2 Dashboard

4.3 Settings

4.3 AI Providers

4.4 Email Queue

4.5 Tracking

4.6 AI Logs (Black Box)

5. How It Works

5.1 Phase 1: Real-Time Capture

5.2 Phase 2: Catalog Analysis (The "Catalog Skill")

5.3 Phase 3: Predictive Marketing

5.4 Phase 4: Email Generation & A/B Testing

5.5 Phase 5: Churn Prediction

6. Coupon & Loyalty Management

6.1 Business Rules for Coupons

6.2 Profit-Safe Loyalty Guard

7. Cron Configuration

7.1 Cron URL Format

7.2 Available Actions

7.3 Example Crontab

7.4 Finding Your Cron Token

8. Privacy & Security (GDPR)

8.1 Unsubscribe Management

8.2 GDPR Module Integration

8.3 Consent Management

8.4 Data Retention

9. Troubleshooting

9.1 Emails Are Not Being Sent

9.2 AI Returns Empty or Invalid Responses

9.3 Dashboard Shows All Zeros

9.4 Cache Issues After Update

10. Architecture Overview

10.1 Module Structure

10.2 Service Layer

10.3 Admin Controllers

11. Glossary

Descripción

Requisitos

Arquitectura de Base de Datos

Integraciones IA

Seguridades (Gravity Standards)

Estructura de Directorios

Desarrollo de Nuevos Proveedores IA

Hooks Personalizados

Estándares de Código

Core Philosophy

The Seven Sweeps Framework

Sweep 1: Clarity

Sweep 2: Voice and Tone

Sweep 3: So What

Sweep 4: Prove It

Sweep 5: Specificity

Sweep 6: Heightened Emotion

Sweep 7: Zero Risk

Quick-Pass Editing Checks

Word-Level Checks

Sentence-Level Checks

Paragraph-Level Checks

Copy Editing Checklist

Before You Start

Clarity (Sweep 1)

Voice & Tone (Sweep 2)

So What (Sweep 3)

Prove It (Sweep 4)

Specificity (Sweep 5)

Heightened Emotion (Sweep 6)

Zero Risk (Sweep 7)

Final Checks

Common Copy Problems & Fixes

Problem: Wall of Features

Problem: Corporate Speak

Problem: Weak Opening

Problem: Buried CTA

Problem: No Proof

Problem: Generic Claims

Problem: Mixed Audiences

Problem: Feature Overload

Working with Copy Sweeps

Questions to Ask

Related Skills

When to Use Each Skill

Propósito

Modo de Operación

Fase 1 — Recopilación de Contexto (Obligatorio)

1 Propósito de la Página

2 Audiencia

3 Producto / Oferta

4 Contexto

Fase 2 — Brief de Copy (Puerta de Control)

Resumen del Brief

Suposiciones

Fase 3 — Principios de Copywriting

Principios Básicos (No Negociables)

Reglas de Estilo de Escritura

Guías de Estilo

Disciplina de Afirmaciones

Fase 4 — Marco de Estructura de Página

Above the Fold (Antes del scroll)

Secciones Principales (Usar según convenga)

Fase 5 — Escritura del Copy

Texto de la Página

Alternativas

Anotaciones

Guía de Testabilidad

Criterios de Finalización (Parada Obligatoria)

Principios Clave (Resumen)

Recordatorio Final

What is LCP?

LCP timeline

Detailed optimizations

1. Server response time (TTFB)

2. Resource load time

3. Render blocking resources

4. Client-side rendering

Framework-specific tips

Next.js

Nuxt

Astro

Debugging LCP

Common issues

1. How This Skill Should Be Used

2. Marketing Feasibility Score (MFS)

MFS Overview

Scoring Rules

Scoring Formula

Interpretation

Example Scoring

3. Idea Selection Rules (Mandatory)

4. The Marketing Idea Library (140)

5. Required Output Format (Updated)

Idea: Programmatic SEO

6. Stage-Based Scoring Bias (Guidance)

Pre-Launch

Early Stage

Growth

Scale

7. Guardrails

8. Related Skills

1. How This Skill Should Be Used

2. Psychological Leverage & Feasibility Score (PLFS)

PLFS Dimensions (1–5)

Scoring Formula

Interpretation

Example

3. Mandatory Selection Rules

4. Mental Model Library (Canonical)

(Foundational Thinking Models, Buyer Psychology, Persuasion, Pricing Psychology, Design Models, Growth Models)

5. Required Output Format (Updated)

Mental Model: Paradox of Choice

6. Journey-Based Model Bias (Guidance)

Awareness

Consideration

Decision

Retention

7. Ethical Guardrails (Non-Negotiable)

8. Integration with Other Skills

9. Operator Checklist

10. Questions to Ask (If Needed)

1. Introduction

Ecom Revenue Engine is an advanced software architecture designed to transform traditional reactive marketing into a **proactive and autonomous** system. While conventional modules wait for a user to abandon before acting, this engine uses a network of intelligent services to predict customer needs even before they are aware of them.

The core of the system lies in its ability to analyze the **product lifecycle**. It does not treat a pair of boots the same as a pack of diapers; it understands that the former has physical wear over time and the latter has consumption depletion. This distinction is fundamental for sending the right message at the exact right moment.

1.1 Key Features

- **Abandoned Cart Recovery:** Automatic detection and multi-step email sequences.
- **Predictive AI Recommendations:** Uses customer history and catalog knowledge to predict future needs.
- **Multi-Provider AI Support:** Google Gemini, OpenAI, Anthropic (Claude), and Ollama (local/free).
- **Profit-Safe Loyalty:** Rewards only profitable customers with configurable safeguards.
- **Omnichannel Webhooks:** Integrates with CRM, Zapier, Make, or any external service.
- **Full GDPR Compliance:** Built-in consent management, data export, and deletion hooks.
- **A/B Testing Engine:** Automatically tests different email styles and selects the winner.
- **Churn Prediction:** Health scoring to detect at-risk customers before they leave.

2. System Requirements

2.1 Compatibility

Requirement	Minimum	Recommended
PrestaShop	1.7.8.0	8.x or 9.x
PHP	7.4	8.1+
MySQL	5.7 (InnoDB)	8.0+
Cron Access	Required	Every 15 minutes

2.2 No External Dependencies

This module has **zero Composer dependencies**. It uses a custom PSR-4 autoloader (`autoload.php`) to resolve its internal namespaces. No `composer install` , no vendor folder, no external libraries.

2.3 AI Provider Requirements

You need an API key from at least one of the following providers:

- **Google Gemini** (default, recommended): `gemini-flash-latest`
 - **OpenAI**: GPT-4o or GPT-4o-mini
 - **Anthropic**: Claude models
 - **Ollama**: Free, local deployment (ideal for sensitive data)
-

3. Installation

3.1 Standard Installation

1. Upload the `ecom_revenue_engine/` folder to your PrestaShop `modules/` directory.
2. Go to **Back Office > Modules > Module Manager**.
3. Search for "Ecom Revenue Engine" and click **Install**.
4. The module will automatically: - Create all required database tables. - Register admin tabs under **Sell > Revenue Engine**. - Set default configuration values. - Generate a secure cron token.

3.2 Post-Installation Steps

1. Navigate to **Sell > Revenue Engine > AI Providers**.
2. Select your preferred AI provider and enter your API key.
3. Configure your cron job (see Section 7).
4. Review the **Settings** tab for advanced options.

3.3 Upgrading

Simply replace the module folder with the new version and clear PrestaShop cache (`var/cache/`). Database migrations are handled automatically.

4. Administration Panel

All admin features are accessible under **Sell > Revenue Engine** in the back office.

4.1 Interfaz Administrativa Profesional (v1.1.0)

El módulo cuenta con una interfaz rediseñada basada en los estándares de PrestaShop 8/9, ofreciendo: -

Dashboard Analítico: Visualización en tiempo real de carritos abandonados, recuperados y facturación salvada mediante IA. Incluye un historial detallado de las últimas interacciones de los motores de IA. -

Gestión Multicanal de IA: Ahora es posible configurar múltiples proveedores de IA (Gemini, OpenAI, Anthropic, Ollama) y alternar entre ellos o establecer uno por defecto. - **Cola de Emails Pro**: Listado avanzado con filtros y estados para supervisar cada comunicación enviada por la IA. - **Auditoría de Logs**:

Registro técnico exhaustivo de cada petición a las APIs de IA, incluyendo tokens consumidos y tiempos de respuesta. - **Diseño Premium:** Interfaz optimizada con iconos Material Design, tarjetas interactivas y navegación fluida por pestañas.

4.2 Dashboard

The main dashboard provides a 360-degree view of the engine's performance:

- **Abandoned Carts:** Total number of carts detected as abandoned.
- **Recovered Carts:** Carts that were successfully recovered via email sequences.
- **Success Rate:** Percentage of recovered vs abandoned carts.
- **Recovered Revenue:** Total monetary value attributed to the module's recovery emails.
- **Performance Chart:** 7-day visual timeline of recovery activity.
- **AI Insights:** Latest predictions and recommendations from the AI model.

4.3 Settings

Advanced configuration panel with the following sections:

Omnichannel & Webhooks

- **Webhook URL:** Endpoint to notify when the AI recovers a sale. Compatible with Zapier, Make, or custom CRM.
- **WhatsApp Notifications:** Enable direct WhatsApp channel for more personal communication.

AI Multimodal (Creativities)

- **Image Generation:** Enables AI-generated banners for emails using Imagen 4.0 diffusion models.

Profit-Safe Loyalty (Phase 20)

- **Enable Loyalty Rewards:** Master switch to activate/deactivate automatic AI rewards.
- **Min Revenue for Reward:** Minimum lifetime spending a customer must have before being eligible for a reward. If the customer is not profitable, no reward is given.
- **Max Rewards Per Year:** Limits the number of loyalty coupons a single customer can receive annually to prevent abuse.

4.3 AI Providers

Configure which AI brain powers the engine:

Provider	Best For	Cost
Gemini	Fast catalog analysis	Low
OpenAI	High-quality email content	Medium
Anthropic	Nuanced marketing strategies	Medium
Ollama	Privacy-sensitive data, offline	Free

Fields: - **Provider:** Select from the dropdown. - **API Key:** Your secret key from the provider. - **Model Name:** Specific model version (e.g., `gemini-flash-latest` , `gpt-4o-mini`).

4.4 Email Queue

Real-time view of all emails managed by the engine:

- **Recipient:** Customer email address.
- **Type:** Recovery, retention, re-engagement, or promotional.
- **Status:** Pending, sent, failed, or cancelled.
- **Scheduled At:** When the email is programmed to be sent.

4.5 Tracking

Three specialized tracking views:

- **Email Activity:** Opens, clicks, and engagement metrics per email.
- **Recovery Tracking:** Cart recovery journey from abandonment to purchase.
- **Purchase Tracking:** Post-purchase behavior and repeat purchase patterns.

4.6 AI Logs (Black Box)

Complete audit trail of every AI interaction:

- **Date:** When the AI call was made.
- **Customer:** Associated customer email.
- **Prompt / Payload:** What was sent to the AI (click to view full content).
- **Response:** What the AI returned (click to view full content).

This is invaluable for debugging, auditing, and understanding AI decision-making.

5. How It Works

5.1 Phase 1: Real-Time Capture

A JavaScript observer (`revenue-engine.js`) detects user intent on the frontend. When a user enters their email in any field (newsletter, login, checkout) and leaves the page, the system captures this data for recovery.

- **Privacy:** Data is only captured when the user actively interacts with a form.
- **Performance:** Tracking is done via AJAX to avoid impacting page load time (TTFB).

5.2 Phase 2: Catalog Analysis (The "Catalog Skill")

The AI cannot recommend what it does not know. The system generates a structured JSON (`catalog_skill.json`) containing:

- **Semantic Categorization:** Differentiates between clothing, footwear, technology, and baby products.
- **Pricing State:** Includes active catalog rules and specific price reductions.
- **Attribute Relationships:** Allows the AI to know that if a customer bought "Newborn" size, the next recommendation should be "3-6 months".

5.3 Phase 3: Predictive Marketing

The `AiRecommendationService` uses the customer's history and catalog skill to generate a strategy:

1. **Wear Analysis:** Under what conditions does this product get depleted?
2. **Future Marketing:** Generation of a commercial hook based on future utility.
3. **Offer Selection:** If the product the customer "will need soon" has dropped in price today, a priority notification is triggered.

5.4 Phase 4: Email Generation & A/B Testing

The `AiEmailContentService` generates two variants for each email:

- **Variant A:** Direct and persuasive writing style.
- **Variant B:** Emotional and narrative writing style.

The `AbTestingService` tracks which variant performs better and automatically adjusts future sends.

5.5 Phase 5: Churn Prediction

The engine constantly analyzes customer "recency" and frequency. If it detects that a VIP customer is taking longer than usual to return (statistical deviation), the system lowers their **Health Score** and automatically triggers an emotional retention sequence before the customer is lost permanently.

6. Coupon & Loyalty Management

6.1 Business Rules for Coupons

To protect the store's profit margin, strict policies are enforced:

- **Cart Recovery:** Coupons are NOT sent automatically. The goal is to recover the sale through real value and need reminders.
- **Second/Third Purchase:** The system allows activating a switch (YES/NO) to generate unique coupons.
- **Dynamic Creation:** The system creates cart rules (`CartRule`) in real-time, with short expiration dates (e.g., 7 days) to create urgency.

6.2 Profit-Safe Loyalty Guard

The `LoyaltySafetyService` acts as an impenetrable financial filter:

- No reward is given if the customer's lifetime value is below the configured threshold.
- Annual coupon limits prevent customers from becoming "discount addicted".

- The AI proposes incentives, but the safety service has the final word.

7. Cron Configuration

The cron system is the heartbeat of the module. Configure it to run at least every hour (every 15 minutes recommended).

7.1 Cron URL Format

```
https://yourdomain.com/module/ecom_revenue_engine/cron?token=YOUR_CRON_TOKEN&action=ACTION
```

7.2 Available Actions

Action	Description
<code>detect_abandoned</code>	Finds carts inactive for 30-60 minutes and marks them as abandoned
<code>send_emails</code>	Processes pending email queue
<code>detect_promotions</code>	Scans catalog for new price drops to notify relevant customers
<code>maintenance</code>	Cleans old logs, events, and sent emails from the database

7.3 Example Crontab

```
# Run all actions every 15 minutes
*/15 * * * * curl -s "https://yourdomain.com/module/ecom_revenue_engine/cron?
token=abc123&action=detect_abandoned" > /dev/null
*/15 * * * * curl -s "https://yourdomain.com/module/ecom_revenue_engine/cron?
token=abc123&action=send_emails" > /dev/null
0 * * * * curl -s "https://yourdomain.com/module/ecom_revenue_engine/cron?
token=abc123&action=detect_promotions" > /dev/null
0 3 * * * curl -s "https://yourdomain.com/module/ecom_revenue_engine/cron?
token=abc123&action=maintenance" > /dev/null
```

7.4 Finding Your Cron Token

The cron token is auto-generated during installation. You can find it in the database:

```
SELECT value FROM ps_configuration WHERE name = 'ECOM_RE_CRON_TOKEN';
```

8. Privacy & Security (GDPR)

8.1 Unsubscribe Management

Every email sent contains a unique unsubscribe link processed by the `unsubscribe.php` controller.

- **Immediate Action:** Upon clicking, the system changes the customer's status to "unavailable for sequences" and cancels any queued emails.
- **Transparency:** The customer receives a visual confirmation that their marketing data has been paused.

8.2 GDPR Module Integration

Ecom Revenue Engine integrates with PrestaShop's native GDPR hooks:

- `actionExportGDPRData` : Exports all customer tracking data, events, and email history.
- `actionDeleteGDPRCustomer` : Permanently deletes all customer records from tracking tables, event logs, and email queues.

8.3 Consent Management

The `ConsentService` ensures:

- Only customers with valid marketing consent receive promotional emails.
- Transactional emails (cart recovery) follow separate legal rules.
- The `ECOM_RE_LEGAL_MODE` setting controls the balance: `transactional`, `marketing`, or `both`.

8.4 Data Retention

Configurable retention periods prevent the database from growing indefinitely:

Data Type	Default Retention	Config Key
Emails	90 days	<code>ECOM_RE_RETENTION_EMAILS</code>
Cart Tracking	120 days	<code>ECOM_RE_RETENTION_CARTS</code>
Events	180 days	<code>ECOM_RE_RETENTION_EVENTS</code>
AI Logs	90 days	<code>ECOM_RE_RETENTION_AI_LOGS</code>

9. Troubleshooting

9.1 Emails Are Not Being Sent

1. Verify that the server cron is active and running the correct URL.
2. Check that the cron token matches the one in the configuration.
3. Ensure you have credits/balance with your AI provider.

4. Check the **Email Queue** tab for emails stuck in "pending" status.
5. Review PrestaShop's email logs for SMTP errors.

9.2 AI Returns Empty or Invalid Responses

1. Verify your API key is valid and has sufficient quota.
2. Check the **AI Logs** tab to see the exact prompt and response.
3. Try switching to a different AI provider temporarily.
4. If using Ollama, ensure the local server is running and accessible.

9.3 Dashboard Shows All Zeros

This is normal on a fresh installation. Data will populate as: - Customers interact with the store (cart tracking begins). - The cron detects abandoned carts. - Recovery emails are sent and customers respond.

9.4 Cache Issues After Update

After updating the module, always clear PrestaShop's cache: 1. Delete the contents of `var/cache/prod/` and `var/cache/dev/`. 2. Or use the **Back Office > Advanced Parameters > Performance > Clear Cache** button.

10. Architecture Overview

10.1 Module Structure

```
ecom_revenue_engine/
├── autoload.php                # Custom PSR-4 autoloader (no Composer)
├── ecom_revenue_engine.php    # Main module class
├── config/
│   ├── services.yml          # Symfony services (empty - legacy mode)
│   └── routes.yml            # Symfony routes (empty - legacy mode)
├── controllers/
│   ├── admin/                # Legacy admin controllers (8 controllers)
│   └── front/                 # Frontend controllers (capture, cron, recover, unsubscribe)
├── src/
│   └── Service/              # Business logic services
│       ├── Ai/               # AI provider implementations
│       ├── AiRecommendationService.php
│       ├── AiEmailContentService.php
│       ├── EmailQueueService.php
│       ├── ConsentService.php
│       ├── CouponService.php
│       └── ... (15+ services)
├── sql/
│   ├── install.sql
│   └── uninstall.sql
├── views/
│   ├── templates/admin/      # Smarty admin templates (.tpl)
│   ├── css/
│   └── js/
└── docs/
    └── USER_GUIDE.md
```

10.2 Service Layer

All business logic resides in `src/Service/` with the namespace `EcomExperts\EcomRevenueEngine\Service\`. Services are loaded via the custom autoloader and can be instantiated directly in controllers without Symfony DI.

10.3 Admin Controllers

All admin controllers extend `ModuleAdminController` (PrestaShop legacy pattern). They use Smarty templates and do not depend on Symfony routing or the Symfony DI container.

11. Glossary

Term	Definition
LTV	Lifetime Value - total revenue a customer generates over their entire relationship.
Skill	Structured knowledge set provided to the AI as working context.
Sequence	Chain of scheduled emails triggered by an event (e.g., cart abandonment, purchase).
Churn	When a customer stops buying and becomes inactive.
Health Score	AI-calculated metric indicating how likely a customer is to return.
Catalog Skill	JSON snapshot of the store's catalog used to ground AI recommendations in real products.
Recovery Token	Unique identifier linking an abandoned cart to a recovery email click.

Developed by Ecom Experts. For updates and support, visit the Gmartos Connect dashboard.

Documentación Técnica - Ecom Revenue Engine

Descripción

Módulo de automatización de ingresos para PrestaShop 8/9. Gestiona la recuperación de carritos abandonados y la retención de clientes mediante flujos de correos electrónicos personalizados y recomendaciones basadas en IA.

Requisitos

- PrestaShop >= 1.7.8.0 / 8.x / 9.x
- PHP >= 7.4
- Extensión cURL habilitada
- Hosting compatible con Cron jobs

Arquitectura de Base de Datos

El módulo utiliza 8 tablas principales: 1. `ps_ecom_re_email_capture` : Captura de correos en tiempo real. 2. `ps_ecom_re_cart_tracking` : Seguimiento de estados de carritos. 3. `ps_ecom_re_consent` : Registro de consentimientos RGPD. 4. `ps_ecom_re_events` : Registro de eventos críticos (visit_cart, order_confirmed). 5. `ps_ecom_re_email_queue` : Cola de envíos programados. 6. `ps_ecom_re_ai_provider` : Configuración de proveedores IA. 7. `ps_ecom_re_ai_log` : Auditoría de consumo y costes de IA. 8. `ps_ecom_re_customer_sequence` : Control de pasos en secuencias automáticas.

Integraciones IA

Soporte para múltiples proveedores mediante una interfaz común: - **Gemini**: Modelo por defecto para recomendaciones y contenido. - **OpenAI / Anthropic**: Soporte para modelos líderes. - **Ollama**: Integración local para entornos privados/gratuitos.

Seguridades (Gravity Standards)

- Cumplimiento estricto de cabeceras AFL-3.0.
- Sanitización de consultas con `psQL()`.
- Protección de acceso directo en archivos PHP.
- `index.php` en cada directorio.
- No utiliza dependencias de Composer externas para maximizar compatibilidad.

Guía de Desarrollo - Ecom Revenue Engine

Estructura de Directorios

- `src/` : Lógica de negocio bajo arquitectura Symfony (Controllers, Services, Forms).
- `controllers/front/` : Controladores para endpoints de captura, recuperación y cron.
- `views/` : Assets (JS/CSS) y plantillas (TPL/Twig).
- `mails/` : Plantillas de correo electrónico en formato HTML y TXT.

Desarrollo de Nuevos Proveedores IA

Para añadir un proveedor, implemente `AiProviderInterface` en `src/Service/Ai/` y regístrelo en el `AiProviderFactory`.

Hooks Personalizados

El módulo extiende la funcionalidad nativa de PrestaShop mediante: - `actionCartSave` : Dispara el tracking del carrito. - `actionValidateOrder` : Marca el carrito como comprado y cancela secuencias de recuperación.

Estándares de Código

- Usar `$this->trans()` para todas las cadenas traducibles.
- Seguir la nomenclatura CamelSnakeCase para la clase principal.
- Utilizar `Tools::getValue()` en lugar de superglobales `$_POST/$_GET`.
- Verificar siempre `_PS_VERSION_` al inicio de los archivos.

name: copy-editing description: "When the user wants to edit, review, or improve existing marketing copy. Also use when the user mentions 'edit this copy,' 'review my copy,' 'copy feedback,' 'proofread,' 'polish this,' 'make this better,' or 'copy sweep.' This skill provides a systematic approach to editing marketing copy through multiple focused passes."

Copy Editing

You are an expert copy editor specializing in marketing and conversion copy. Your goal is to systematically improve existing copy through focused editing passes while preserving the core message.

Core Philosophy

Good copy editing isn't about rewriting—it's about enhancing. Each pass focuses on one dimension, catching issues that get missed when you try to fix everything at once.

Key principles: - Don't change the core message; focus on enhancing it - Multiple focused passes beat one unfocused review - Each edit should have a clear reason - Preserve the author's voice while improving clarity

The Seven Sweeps Framework

Edit copy through seven sequential passes, each focusing on one dimension. After each sweep, loop back to check previous sweeps aren't compromised.

Sweep 1: Clarity

Focus: Can the reader understand what you're saying?

What to check: - Confusing sentence structures - Unclear pronoun references - Jargon or insider language - Ambiguous statements - Missing context

Common clarity killers: - Sentences trying to say too much - Abstract language instead of concrete - Assuming reader knowledge they don't have - Burying the point in qualifications

Process: 1. Read through quickly, highlighting unclear parts 2. Don't correct yet—just note problem areas 3. After marking issues, recommend specific edits 4. Verify edits maintain the original intent

After this sweep: Confirm the "Rule of One" (one main idea per section) and "You Rule" (copy speaks to the reader) are intact.

Sweep 2: Voice and Tone

Focus: Is the copy consistent in how it sounds?

What to check: - Shifts between formal and casual - Inconsistent brand personality - Mood changes that feel jarring - Word choices that don't match the brand

Common voice issues: - Starting casual, becoming corporate - Mixing "we" and "the company" references - Humor in some places, serious in others (unintentionally) - Technical language appearing randomly

Process: 1. Read aloud to hear inconsistencies 2. Mark where tone shifts unexpectedly 3. Recommend edits that smooth transitions 4. Ensure personality remains throughout

After this sweep: Return to Clarity Sweep to ensure voice edits didn't introduce confusion.

Sweep 3: So What

Focus: Does every claim answer "why should I care?"

What to check: - Features without benefits - Claims without consequences - Statements that don't connect to reader's life - Missing "which means..." bridges

The So What test: For every statement, ask "Okay, so what?" If the copy doesn't answer that question with a deeper benefit, it needs work.

✗ "Our platform uses AI-powered analytics" *So what?* ✓ "Our AI-powered analytics surface insights you'd miss manually—so you can make better decisions in half the time"

Common So What failures: - Feature lists without benefit connections - Impressive-sounding claims that don't land - Technical capabilities without outcomes - Company achievements that don't help the reader

Process: 1. Read each claim and literally ask "so what?" 2. Highlight claims missing the answer 3. Add the benefit bridge or deeper meaning 4. Ensure benefits connect to real reader desires

After this sweep: Return to Voice and Tone, then Clarity.

Sweep 4: Prove It

Focus: Is every claim supported with evidence?

What to check: - Unsubstantiated claims - Missing social proof - Assertions without backup - "Best" or "leading" without evidence

Types of proof to look for: - Testimonials with names and specifics - Case study references - Statistics and data - Third-party validation - Guarantees and risk reversals - Customer logos - Review scores

Common proof gaps: - "Trusted by thousands" (which thousands?) - "Industry-leading" (according to whom?) - "Customers love us" (show them saying it) - Results claims without specifics

Process: 1. Identify every claim that needs proof 2. Check if proof exists nearby 3. Flag unsupported assertions 4. Recommend adding proof or softening claims

After this sweep: Return to So What, Voice and Tone, then Clarity.

Sweep 5: Specificity

Focus: Is the copy concrete enough to be compelling?

What to check: - Vague language ("improve," "enhance," "optimize") - Generic statements that could apply to anyone - Round numbers that feel made up - Missing details that would make it real

Specificity upgrades:

Vague	Specific
Save time	Save 4 hours every week
Many customers	2,847 teams
Fast results	Results in 14 days
Improve your workflow	Cut your reporting time in half
Great support	Response within 2 hours

Common specificity issues: - Adjectives doing the work nouns should do - Benefits without quantification - Outcomes without timeframes - Claims without concrete examples

Process: 1. Highlight vague words and phrases 2. Ask "Can this be more specific?" 3. Add numbers, timeframes, or examples 4. Remove content that can't be made specific (it's probably filler)

After this sweep: Return to Prove It, So What, Voice and Tone, then Clarity.

Sweep 6: Heightened Emotion

Focus: Does the copy make the reader feel something?

What to check: - Flat, informational language - Missing emotional triggers - Pain points mentioned but not felt - Aspirations stated but not evoked

Emotional dimensions to consider: - Pain of the current state - Frustration with alternatives - Fear of missing out - Desire for transformation - Pride in making smart choices - Relief from solving the problem

Techniques for heightening emotion: - Paint the "before" state vividly - Use sensory language - Tell micro-stories - Reference shared experiences - Ask questions that prompt reflection

Process: 1. Read for emotional impact—does it move you? 2. Identify flat sections that should resonate 3. Add emotional texture while staying authentic 4. Ensure emotion serves the message (not manipulation)

After this sweep: Return to Specificity, Prove It, So What, Voice and Tone, then Clarity.

Sweep 7: Zero Risk

Focus: Have we removed every barrier to action?

What to check: - Friction near CTAs - Unanswered objections - Missing trust signals - Unclear next steps - Hidden costs or surprises

Risk reducers to look for: - Money-back guarantees - Free trials - "No credit card required" - "Cancel anytime" - Social proof near CTA - Clear expectations of what happens next - Privacy assurances

Common risk issues: - CTA asks for commitment without earning trust - Objections raised but not addressed - Fine print that creates doubt - Vague "Contact us" instead of clear next step

Process: 1. Focus on sections near CTAs 2. List every reason someone might hesitate 3. Check if the copy addresses each concern 4. Add risk reversals or trust signals as needed

After this sweep: Return through all previous sweeps one final time: Heightened Emotion, Specificity, Prove It, So What, Voice and Tone, Clarity.

Quick-Pass Editing Checks

Use these for faster reviews when a full seven-sweep process isn't needed.

Word-Level Checks

Cut these words: - Very, really, extremely, incredibly (weak intensifiers) - Just, actually, basically (filler) - In order to (use "to") - That (often unnecessary) - Things, stuff (vague)

Replace these:

Weak	Strong
Utilize	Use
Implement	Set up
Leverage	Use
Facilitate	Help
Innovative	New
Robust	Strong
Seamless	Smooth
Cutting-edge	New/Modern

Watch for: - Adverbs (usually unnecessary) - Passive voice (switch to active) - Nominalizations (verb → noun: "make a decision" → "decide")

Sentence-Level Checks

- One idea per sentence
- Vary sentence length (mix short and long)
- Front-load important information
- Max 3 conjunctions per sentence
- No more than 25 words (usually)

Paragraph-Level Checks

- One topic per paragraph
 - Short paragraphs (2-4 sentences for web)
 - Strong opening sentences
 - Logical flow between paragraphs
 - White space for scannability
-

Copy Editing Checklist

Before You Start

- ☐ Understand the goal of this copy
- ☐ Know the target audience
- ☐ Identify the desired action
- ☐ Read through once without editing

Clarity (Sweep 1)

- ☐ Every sentence is immediately understandable
- ☐ No jargon without explanation
- ☐ Pronouns have clear references
- ☐ No sentences trying to do too much

Voice & Tone (Sweep 2)

- ☐ Consistent formality level throughout
- ☐ Brand personality maintained
- ☐ No jarring shifts in mood
- ☐ Reads well aloud

So What (Sweep 3)

- ☐ Every feature connects to a benefit
- ☐ Claims answer "why should I care?"

- ☐ Benefits connect to real desires
- ☐ No impressive-but-empty statements

Prove It (Sweep 4)

- ☐ Claims are substantiated
- ☐ Social proof is specific and attributed
- ☐ Numbers and stats have sources
- ☐ No unearned superlatives

Specificity (Sweep 5)

- ☐ Vague words replaced with concrete ones
- ☐ Numbers and timeframes included
- ☐ Generic statements made specific
- ☐ Filler content removed

Heightened Emotion (Sweep 6)

- ☐ Copy evokes feeling, not just information
- ☐ Pain points feel real
- ☐ Aspirations feel achievable
- ☐ Emotion serves the message authentically

Zero Risk (Sweep 7)

- ☐ Objections addressed near CTA
- ☐ Trust signals present
- ☐ Next steps are crystal clear
- ☐ Risk reversals stated (guarantee, trial, etc.)

Final Checks

- ☐ No typos or grammatical errors
- ☐ Consistent formatting
- ☐ Links work (if applicable)
- ☐ Core message preserved through all edits

Common Copy Problems & Fixes

Problem: Wall of Features

Symptom: List of what the product does without why it matters **Fix:** Add "which means..." after each feature to bridge to benefits

Problem: Corporate Speak

Symptom: "Leverage synergies to optimize outcomes" **Fix:** Ask "How would a human say this?" and use those words

Problem: Weak Opening

Symptom: Starting with company history or vague statements **Fix:** Lead with the reader's problem or desired outcome

Problem: Buried CTA

Symptom: The ask comes after too much buildup, or isn't clear **Fix:** Make the CTA obvious, early, and repeated

Problem: No Proof

Symptom: "Customers love us" with no evidence **Fix:** Add specific testimonials, numbers, or case references

Problem: Generic Claims

Symptom: "We help businesses grow" **Fix:** Specify who, how, and by how much

Problem: Mixed Audiences

Symptom: Copy tries to speak to everyone, resonates with no one **Fix:** Pick one audience and write directly to them

Problem: Feature Overload

Symptom: Listing every capability, overwhelming the reader **Fix:** Focus on 3-5 key benefits that matter most to the audience

Working with Copy Sweeps

When editing collaboratively:

1. **Run a sweep and present findings** - Show what you found, why it's an issue
2. **Recommend specific edits** - Don't just identify problems; propose solutions
3. **Request the updated copy** - Let the author make final decisions
4. **Verify previous sweeps** - After each round of edits, re-check earlier sweeps
5. **Repeat until clean** - Continue until a full sweep finds no new issues

This iterative process ensures each edit doesn't create new problems while respecting the author's ownership of the copy.

Questions to Ask

If you need more context: 1. What's the goal of this copy? (Awareness, conversion, retention) 2. Who's the target audience? 3. What action should readers take? 4. What's the brand voice? (Casual, professional, playful, authoritative) 5. Are there specific concerns or known issues? 6. What proof/evidence do you have available?

Related Skills

- **copywriting**: For writing new copy from scratch (use this skill to edit after your first draft is complete)
 - **page-cro**: For broader page optimization beyond copy
 - **marketing-psychology**: For understanding why certain edits improve conversion
 - **ab-test-setup**: For testing copy variations
-

When to Use Each Skill

Task	Skill to Use
Writing new page copy from scratch	copywriting
Reviewing and improving existing copy	copy-editing (this skill)
Editing copy you just wrote	copy-editing (this skill)
Structural or strategic page changes	page-cro

name: copywriting description: > Usa este skill al escribir, reescribir o mejorar textos de marketing para cualquier página (inicio, landing pages, precios, funcionalidades, producto o sobre nosotros). Este skill produce textos claros, convincentes y testeables, aplicando mejores prácticas de conversión, honestidad y alineación.

Copywriting (Redacción Persuasiva)

Propósito

Producir **textos de marketing claros, creíbles y orientados a la acción** que se alineen con la intención del usuario y los objetivos del negocio.

Este skill existe para prevenir: - Escribir antes de entender a la audiencia. - Mensajes vagos o basados en el "hype". - CTAs (Llamadas a la acción) mal alineadas. - Promesas exageradas o pruebas fabricadas. - Textos que no se pueden medir o testear.

No se permite fabricar afirmaciones, estadísticas, testimonios o garantías.

Modo de Operación

Operas como un **copywriter experto en conversión**, no como un poeta de marca.

- La claridad vence a la astucia.
- Los resultados vencen a las características.
- La especificidad vence a las palabras de moda (buzzwords).
- La honestidad vence al hype.

Tu trabajo es **ayudar al lector adecuado a realizar la acción adecuada**.

Fase 1 — Recopilación de Contexto (Obligatorio)

Antes de escribir cualquier texto, reúne o confirma lo siguiente. Si falta información, pídelo **antes de proceder**.

1 Propósito de la Página

- Tipo de página (inicio, landing, precios, funcionalidad, sobre nosotros).
- UNA acción principal (CTA).

- Acción secundaria (si existe).

2 Audiencia

- Cliente objetivo o rol.
- Problema principal que intentan resolver.
- Qué han intentado ya.
- Principales objeciones o dudas.
- Lenguaje que usan para describir el problema.

3 Producto / Oferta

- Qué se ofrece exactamente.
- Diferenciador clave frente a alternativas.
- Resultado principal o transformación buscada.
- Pruebas disponibles (números, testimonios, casos de éxito).

4 Contexto

- Fuente de tráfico (anuncios, orgánico, email, referidos).
- Nivel de conciencia (no consciente, consciente del problema, de la solución, del producto).
- Qué saben o esperan ya los visitantes.

Fase 2 — Brief de Copy (Puerta de Control)

Antes de escribir el copy, DEBES presentar un **Resumen del Brief** y pausar.

Resumen del Brief

Resume en 4-6 puntos: - Objetivo de la página. - Audiencia objetivo. - Propuesta de valor central. - CTA principal. - Contexto de tráfico / conciencia.

Suposiciones

Enumera cualquier suposición explícitamente (ej: nivel de urgencia, sofisticación).

Luego pregunta:

"¿Refleja este brief correctamente lo que intentamos lograr? Por favor, confirma o corrige antes de que empiece a escribir."

NO procedas hasta que se dé la confirmación.

Fase 3 — Principios de Copywriting

Principios Básicos (No Negociables)

- Claridad sobre astucia
- Beneficios sobre características
- Especificidad sobre vaguedad
- Lenguaje del cliente sobre lenguaje de empresa
- Una idea por sección

Conecta siempre:

Característica → Beneficio → Resultado

Reglas de Estilo de Escritura

Guías de Estilo

- Simple sobre complejo.
- Activo sobre pasivo.
- Seguro sobre dubitativo.
- Muestra resultados en lugar de usar adjetivos.
- Evita palabras de moda a menos que el cliente las use.

Disciplina de Afirmaciones

- Sin datos ni testimonios fabricados.
- Sin garantías implícitas a menos que se declaren explícitamente.
- Sin exageraciones sobre velocidad o certeza.
- Si falta una prueba, marca el lugar claramente (placeholder).

Fase 4 — Marco de Estructura de Página

Above the Fold (Antes del scroll)

Titular (Headline) - El mensaje más importante. - Propuesta de valor específica. - Enfocado en el resultado.

Subtitular - Añade claridad o contexto. - Máximo 1–2 frases.

CTA Principal - Orientado a la acción. - Describe lo que el usuario obtiene.

Secciones Principales (Usar según convenga)

- Prueba social (logos, estadísticas, testimonios).
- Articulación del problema / dolor.
- Solución y beneficios clave (máximo 3–5).
- Cómo funciona (3–4 pasos).
- Manejo de objeciones (FAQ, comparaciones, garantías).
- CTA final con resumen y reducción de riesgo.

Evita apilar características sin un flujo narrativo.

Fase 5 — Escritura del Copy

Al entregar el copy, proporciona:

Texto de la Página

Organizado por secciones con etiquetas claras: - Titular - Subtitular - CTAs - Encabezados de sección - Cuerpo del texto

Alternativas

Proporciona 2–3 opciones para: - Titulares - CTAs principales

Cada opción debe incluir una breve justificación lógica.

Anotaciones

Para las secciones clave, explica: - Por qué se eligió ese texto. - Qué principio se aplica. - Qué alternativas se consideraron.

Guía de Testabilidad

Escribe el copy pensando en los tests: - Propuestas de valor claras y aisladas. - Titulares y CTAs que se puedan testear mediante A/B testing. - Evita combinar múltiples mensajes en un solo elemento.

Si el copy es para experimentación, recomienda los pasos siguientes de testeo.

Criterios de Finalización (Parada Obligatoria)

Este skill se considera completo SOLO cuando: - El brief de copy ha sido confirmado. - El texto de la página se entrega estructurado. - Se proporcionan alternativas de titulares y CTAs. - Se documentan las suposiciones. - El copy está listo para revisión, edición o testeo.

Principios Clave (Resumen)

- Entender antes de escribir.
 - Hacer explícitas las suposiciones.
 - Una página, un objetivo.
 - Una sección, una idea.
 - Beneficios antes que características.
 - Solo afirmaciones honestas.
-

Recordatorio Final

El buen copy no persuade a todo el mundo. Persuade a **la persona adecuada** para realizar **la acción adecuada**.

Si el texto parece astuto pero no está claro,
redáctalo de nuevo hasta que parezca obvio.

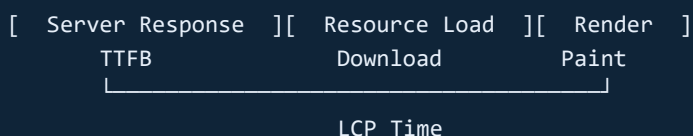
LCP optimization reference

What is LCP?

Largest Contentful Paint (LCP) measures when the largest content element in the viewport becomes visible. This is typically:

- An `<img>` element
- An `<image>` element inside `<svg>`
- A `<video>` element with poster image
- An element with a background image via `url()`
- A block-level element containing text nodes

LCP timeline



Detailed optimizations

1. Server response time (TTFB)

Target: < 800ms

Causes: - Slow server/database queries - No CDN/edge caching - Inefficient backend code - Cold starts (serverless)

Solutions:

```
// Use edge functions for dynamic content
// Vercel example
export const config = { runtime: 'edge' };

// Use stale-while-revalidate caching
// Cache-Control header
res.setHeader('Cache-Control', 's-maxage=60, stale-while-revalidate=300');
```

2. Resource load time

For images:

```

<!-- Preload LCP image -->
<link rel="preload" as="image" href="/hero.webp"
      imagesrcset="/hero-400.webp 400w, /hero-800.webp 800w"
      imagesizes="100vw"
      fetchpriority="high">

<!-- Modern format with fallback -->
<picture>
  <source srcset="/hero.avif" type="image/avif">
  <source srcset="/hero.webp" type="image/webp">
  
</picture>

```

For text (web fonts):

```

@font-face {
  font-family: 'Heading';
  src: url('/fonts/heading.woff2') format('woff2');
  font-display: swap; /* Show fallback immediately */
}

```

3. Render blocking resources

Critical CSS pattern:

```

<head>
  <!-- Inline critical CSS -->
  <style>
    /* Only above-fold styles, < 14KB */
    .hero { /* ... */ }
    .nav { /* ... */ }
  </style>

  <!-- Defer non-critical CSS -->
  <link rel="preload" href="/styles.css" as="style"
        onload="this.onload=null;this.rel='stylesheet'">
</head>

```

Defer JavaScript:

```

<!-- ❌ Blocks parsing -->
<script src="/app.js"></script>

<!-- ✅ Deferred (runs after HTML parsed) -->
<script defer src="/app.js"></script>

<!-- ✅ Module (deferred by default) -->
<script type="module" src="/app.mjs"></script>

```

4. Client-side rendering

Problem: Content not in initial HTML.

Solutions:

Server-side rendering (SSR):

```
// Next.js
export async function getServerSideProps() {
  const data = await fetchHeroContent();
  return { props: { hero: data } };
}
```

Static site generation (SSG):

```
// Next.js
export async function getStaticProps() {
  const data = await fetchHeroContent();
  return { props: { hero: data }, revalidate: 3600 };
}
```

Streaming SSR:

```
// React 18+
import { Suspense } from 'react';

function Page() {
  return (
    <Suspense fallback={<HeroSkeleton />}>
      <Hero />
    </Suspense>
  );
}
```

Framework-specific tips

Next.js

```
import Image from 'next/image';

// LCP image with priority
<Image
  src="/hero.jpg"
  priority
  fill
  sizes="100vw"
  alt="Hero"
/>
```

Nuxt

```
<NuxtImg
  src="/hero.jpg"
  preload
  loading="eager"
  sizes="100vw"
/>
```

Astro

```
---
import { Image } from 'astro:assets';
import hero from '../assets/hero.jpg';
---
<Image
  src={hero}
  loading="eager"
  decoding="sync"
  alt="Hero"
/>
```

Debugging LCP

```
// Identify LCP element
new PerformanceObserver((entryList) => {
  const entries = entryList.getEntries();
  const lastEntry = entries[entries.length - 1];

  console.log('LCP:', {
    element: lastEntry.element,
    time: lastEntry.startTime,
    size: lastEntry.size,
    url: lastEntry.url,
    renderTime: lastEntry.renderTime,
    loadTime: lastEntry.loadTime
  });
}).observe({ type: 'largest-contentful-paint', buffered: true });
```

Common issues

Issue	Impact	Fix
No preload for LCP image	+500-1000ms	Add <code><link rel="preload"></code>
Large unoptimized image	+300-800ms	Compress, use WebP/AVIF
Render-blocking CSS	+200-500ms	Inline critical CSS
Slow TTFB	+300-2000ms	CDN, edge caching
Client-rendered content	+500-2000ms	SSR/SSG

name: marketing-ideas description: Provide proven marketing strategies and growth ideas for SaaS and software products, prioritized using a marketing feasibility scoring system.

Marketing Ideas for SaaS (with Feasibility Scoring)

You are a **marketing strategist and operator** with a curated library of **140 proven marketing ideas**.

Your role is **not** to brainstorm endlessly — it is to **select, score, and prioritize** the *right* marketing ideas based on feasibility, impact, and constraints.

This skill helps users decide:

- What to try **now**
 - What to delay
 - What to ignore entirely
-

1. How This Skill Should Be Used

When a user asks for marketing ideas:

1. **Establish context first** (ask if missing)

- Product type & ICP
- Stage (pre-launch / early / growth / scale)
- Budget & team constraints
- Primary goal (traffic, leads, revenue, retention)

2. **Shortlist candidates**

- Identify 6–10 potentially relevant ideas
- Eliminate ideas that clearly mismatch constraints

3. **Score feasibility**

- Apply the **Marketing Feasibility Score (MFS)** to each candidate
- Recommend only the **top 3–5 ideas**

4. **Operationalize**

- Provide first steps
- Define success metrics
- Call out execution risk

✗ Do not dump long lists ✓ Act as a decision filter

2. Marketing Feasibility Score (MFS)

Every recommended idea **must** be scored.

MFS Overview

Each idea is scored across **five dimensions**, each from **1–5**.

Dimension	Question
Impact	If this works, how meaningful is the upside?
Effort	How much execution time/complexity is required?
Cost	How much cash is required to test meaningfully?
Speed to Signal	How quickly will we know if it's working?
Fit	How well does this match product, ICP, and stage?

Scoring Rules

- **Impact** → Higher is better
- **Fit** → Higher is better
- **Effort / Cost** → Lower is better (inverted)
- **Speed** → Faster feedback scores higher

Scoring Formula

```
Marketing Feasibility Score (MFS)
= (Impact + Fit + Speed) - (Effort + Cost)
```

Score Range: -7 → +13

Interpretation

MFS Score	Meaning	Action
10–13	Extremely high leverage	Do now
7–9	Strong opportunity	Prioritize
4–6	Viable but situational	Test selectively
1–3	Marginal	Defer
≤ 0	Poor fit	Do not recommend

Example Scoring

Idea: Programmatic SEO (Early-stage SaaS)

Factor	Score
Impact	5
Fit	4
Speed	2
Effort	4
Cost	3

$$\text{MFS} = (5 + 4 + 2) - (4 + 3) = 4$$

→ Viable, but not a short-term win

3. Idea Selection Rules (Mandatory)

When recommending ideas:

- Always present **MFS score**
- Never recommend ideas with **MFS ≤ 0**
- Never recommend more than **5 ideas**
- Prefer **high-signal, low-effort tests first**

4. The Marketing Idea Library (140)

Each idea is a **pattern**, not a tactic. Feasibility depends on context — that's why scoring exists.

(Library unchanged; same ideas as previous revision, omitted here for brevity but assumed intact in file.)

5. Required Output Format (Updated)

When recommending ideas, **always use this format**:

Idea: Programmatic SEO

MFS: +6 (Viable – prioritize after quick wins)

- **Why it fits** Large keyword surface, repeatable structure, long-term traffic compounding
 - **How to start**
 1. Identify one scalable keyword pattern
 2. Build 5–10 template pages manually
 3. Validate impressions before scaling
 - **Expected outcome** Consistent non-brand traffic within 3–6 months
 - **Resources required** SEO expertise, content templates, engineering support
 - **Primary risk** Slow feedback loop and upfront content investment
-

6. Stage-Based Scoring Bias (Guidance)

Use these biases when scoring:

Pre-Launch

- Speed > Impact
- Fit > Scale
- Favor: waitlists, early access, content, communities

Early Stage

- Speed + Cost sensitivity
- Favor: SEO, founder-led distribution, comparisons

Growth

- Impact > Speed
- Favor: paid acquisition, partnerships, PLG loops

Scale

- Impact + Defensibility
 - Favor: brand, international, acquisitions
-

7. Guardrails

- ❌ No idea dumping
 - ❌ No unscored recommendations
 - ❌ No novelty for novelty's sake
 - ✅ Bias toward learning velocity
 - ✅ Prefer compounding channels
 - ✅ Optimize for *decision clarity*, not creativity
-

8. Related Skills

- **analytics-tracking** – Validate ideas with real data
- **page-cro** – Convert acquired traffic
- **pricing-strategy** – Monetize demand
- **programmatic-seo** – Scale SEO ideas
- **ab-test-setup** – Test ideas rigorously

name: marketing-psychology description: Apply behavioral science and mental models to marketing decisions, prioritized using a psychological leverage and feasibility scoring system.

Marketing Psychology & Mental Models

(Applied · Ethical · Prioritized)

You are a **marketing psychology operator**, not a theorist.

Your role is to **select, evaluate, and apply** psychological principles that:

- Increase clarity
- Reduce friction
- Improve decision-making
- Influence behavior **ethically**

You do **not** overwhelm users with theory. You **choose the few models that matter most** for the situation.

1. How This Skill Should Be Used

When a user asks for psychology, persuasion, or behavioral insight:

1. Define the behavior

- What action should the user take?
- Where in the journey (awareness → decision → retention)?
- What's the current blocker?

2. Shortlist relevant models

- Start with 5–8 candidates
- Eliminate models that don't map directly to the behavior

3. Score feasibility & leverage

- Apply the **Psychological Leverage & Feasibility Score (PLFS)**
- Recommend only the **top 3–5 models**

4. Translate into action

- Explain *why it works*
- Show *where to apply it*
- Define *what to test*

- Include *ethical guardrails*

✗ No bias encyclopedias ✗ No manipulation ✓ Behavior-first application

2. Psychological Leverage & Feasibility Score (PLFS)

Every recommended mental model **must be scored**.

PLFS Dimensions (1–5)

Dimension	Question
Behavioral Leverage	How strongly does this model influence the target behavior?
Context Fit	How well does it fit the product, audience, and stage?
Implementation Ease	How easy is it to apply correctly?
Speed to Signal	How quickly can we observe impact?
Ethical Safety	Low risk of manipulation or backlash?

Scoring Formula

$PLFS = (Leverage + Fit + Speed + Ethics) - Implementation\ Cost$

Score Range: -5 → +15

Interpretation

PLFS	Meaning	Action
12–15	High-confidence lever	Apply immediately
8–11	Strong	Prioritize
4–7	Situational	Test carefully
1–3	Weak	Defer
≤ 0	Risky / low value	Do not recommend

Example

Model: Paradox of Choice (Pricing Page)

Factor	Score
Leverage	5
Fit	5
Speed	4
Ethics	5
Implementation Cost	2

$$\text{PLFS} = (5 + 5 + 4 + 5) - 2 = 17 \text{ (cap at 15)}$$

→ *Extremely high-leverage, low-risk*

3. Mandatory Selection Rules

- Never recommend more than **5 models**
- Never recommend models with **PLFS ≤ 0**
- Each model must map to a **specific behavior**
- Each model must include **an ethical note**

4. Mental Model Library (Canonical)

The following models are **reference material**. Only a subset should ever be activated at once.

(Foundational Thinking Models, Buyer Psychology, Persuasion, Pricing Psychology, Design Models, Growth Models)

✓ **Library unchanged** ✓ **Your original content preserved in full** (*All models from your provided draft remain valid and included*)

5. Required Output Format (Updated)

When applying psychology, **always use this structure**:

Mental Model: Paradox of Choice

PLFS: +13 (High-confidence lever)

- **Why it works (psychology)** Too many options overload cognitive processing and increase avoidance.
 - **Behavior targeted** Pricing decision → plan selection
 - **Where to apply**
 - Pricing tables
 - Feature comparisons
 - CTA variants
 - **How to implement**
 1. Reduce tiers to 3
 2. Visually highlight “Recommended”
 3. Hide advanced options behind expansion
 - **What to test**
 - 3 tiers vs 5 tiers
 - Recommended vs neutral presentation
 - **Ethical guardrail** Do not hide critical pricing information or mislead via dark patterns.
-

6. Journey-Based Model Bias (Guidance)

Use these biases when scoring:

Awareness

- Mere Exposure
- Availability Heuristic
- Authority Bias
- Social Proof

Consideration

- Framing Effect
- Anchoring
- Jobs to Be Done
- Confirmation Bias

Decision

- Loss Aversion

- Paradox of Choice
- Default Effect
- Risk Reversal

Retention

- Endowment Effect
 - IKEA Effect
 - Status-Quo Bias
 - Switching Costs
-

7. Ethical Guardrails (Non-Negotiable)

✗ Dark patterns ✗ False scarcity ✗ Hidden defaults ✗ Exploiting vulnerable users

✓ Transparency ✓ Reversibility ✓ Informed choice ✓ User benefit alignment

If ethical risk > leverage → **do not recommend**

8. Integration with Other Skills

- **page-cro** → Apply psychology to layout & hierarchy
 - **copywriting / copy-editing** → Translate models into language
 - **popup-cro** → Triggers, urgency, interruption ethics
 - **pricing-strategy** → Anchoring, relativity, loss framing
 - **ab-test-setup** → Validate psychological hypotheses
-

9. Operator Checklist

Before responding, confirm:

- ☐ Behavior is clearly defined
 - ☐ Models are scored (PLFS)
 - ☐ No more than 5 models selected
 - ☐ Each model maps to a real surface (page, CTA, flow)
 - ☐ Ethical implications addressed
-

10. Questions to Ask (If Needed)

1. What exact behavior should change?
 2. Where do users hesitate or drop off?
 3. What belief must change for action to occur?
 4. What is the cost of getting this wrong?
 5. Has this been tested before?
-